

Observation Report

1. Test Purpose and Method

The purpose of this observation was to understand users' natural behavior, operational difficulties, and subjective feelings when interacting with the electronic pet prototype. The think-aloud method was used, where participants spoke their thoughts aloud while operating the device. An observer recorded key actions, moments of hesitation, error attempts, and spontaneous verbal comments. The test environments included quiet offline classrooms and dormitories, as well as online remote observation via WeChat. Each participant interacted for approximately 15 minutes.

2. Participant Backgrounds

Among the five participants, two were recommended by friends, one was a dormitory roommate of a team member, and two were students from another course project group. Two participants were tested in person (offline), and three were tested online.

3. Observation Records

Initial Contact and Wake-Up

Without any prompts, the first action of every participant upon receiving the electronic pet was to look at the screen or flip the device over to examine it as a whole. In the offline test, P1 picked up the device, looked around its edges, then naturally touched the screen with a finger. After seeing the facial expression change on the screen, P1 continued swiping. In the dormitory, P3 stared at the screen for about twenty seconds without actively touching it and instead asked directly, "How do you play with this?" Only after receiving a slight hint did P3 start touching the screen. In the online test, P4 held the device up to the camera for a long time and then said, "Is it listening? There's no sign." Throughout the entire process, no participant actively tried to speak to the device or noticed the voice module nearby; everyone focused their attention on the screen first. This indicates that without visual guidance, the voice function was completely overlooked.

Expression Interaction and Feedback

As participants began to touch or tap the pet repeatedly, expression changes gradually became the focus. P2 saw a smile on the first touch, dizziness on the second, and a smile again on the third. P2 tried a fourth time and found the sequence repeating, then said flatly, "It just switches between a few fixed faces. It doesn't feel like a real pet." After two consecutive strokes, P4 also said, "I got bored on the second try." After experiencing it for a few minutes, P5 proactively suggested, "Maybe more expressions could be added." The observer noticed that after three or four repetitive operations, most participants' interaction frequency dropped noticeably. Some started looking at their phones, and some stopped to ask, "Are there any

other functions?" The predictability of the expression feedback caused participants to quickly lose interest in further exploration.

Voice Interaction Attempts

After being informed that the device had a voice function (but not told the specific wake word), the participants' performance showed clear confusion. P4 said to the device, "What's the weather like?" , no response. P4 raised their voice slightly and tried again , still no response. P4 brought the device closer and asked a third time; this time it succeeded, but P4 commented, "It's not very stable." P5 also tried to check the weather by voice, failing the first two times and succeeding on the third, but with a long delay. While waiting, P4 said, "I don't know when I can talk to it. Sometimes it responds, sometimes it doesn't react at all." P3, observed online, didn't try voice much but said afterwards, "It would be better if it could be smarter." The observer recorded a common behavior among all participants: before attempting voice, they would all quiet down, bring their face close to the device, and then uncertainly utter a short phrase , not a single participant succeeded on the first try, and no one knew when the device was listening. This uncertainty meant the voice function was almost never used proactively.

Movement and Physical Actions

Regarding the device's motion responses, participants' experiences varied considerably. P1 gently shook the device the first time, and the screen switched to a dizzy expression, which P1 thought made sense. But shaking it a second time produced a smile expression. P1 shook their head and said, "Sometimes it reacts when you shake it, sometimes not. I'm not sure what it's trying to express." When P3 picked up the device and accidentally bumped the table, the device jolted. P3 immediately asked, "Why did it do that?" and then tried actively bumping the table, but the same reaction did not occur. The observer noted that users tended to interpret every movement of the device as some kind of emotional expression. When there was no stable, understandable connection between movement and emotional state, confusion and even disappointment arose.

Problems Caused by Network Dependency

During tests with changing network conditions, problems were exposed very directly. P2 connected to WiFi requiring web authentication in a classroom, and the pet was unable to access the weather API. P2 tried asking for the weather, but the voice assistant did not respond. Somewhat helplessly, P2 said, "In a classroom without WiFi, you can't check the weather, and the voice assistant goes mute." P4 simulated a poor signal scenario; the pet entered the weather screen and froze there for over fifteen seconds before timing out and exiting. During the wait, P4 tried touching the screen to cancel, but there was no response. After the experience, all participants agreed that once disconnected from WiFi, the two core functions , weather and voice , were completely unusable, leaving only expressions and temperature/humidity display. The product's appeal dropped by more than half.

Appearance and Overall Impression

Appearance was the aspect participants noticed earliest and commented on most directly. Within five seconds of picking up the device, P3 said, "I think the appearance is a bit rough; I feel there is still room for improvement." After using it, P1 said seriously, "The whole product's appearance isn't a unified whole; the components are all exposed," then added, "I hope, if possible, you can create a visually appealing look, because I am drawn to many products by their appearance, and I'd like to have more choices of virtual looks." P5 also succinctly commented at the end, "The appearance is not yet attractive enough." The observer noted that participants' first impressions of the product's functionality were often influenced beforehand by its appearance , the rough casing and exposed modules led some users to form the judgment of "this doesn't look like a finished product" before they even experienced the functions.

4. Summary of Main Behavioral Patterns

Synthesizing the performance of the five participants, several clear behavioral patterns can be identified. First, upon first contact, everyone examined the appearance, then touched the screen; very few actively tried voice. Second, faced with repetitive and predictable expression feedback, participants' willingness to interact dropped rapidly after three or four operations. Third, when voice interaction lacked visual status cues, users would repeatedly try different ways of speaking until they either succeeded or gave up, showing a clear sense of uncertainty during the process. Fourth, when the network was disconnected, users would repeatedly attempt to request weather or voice functions, and after multiple failures, would stop using those features. Fifth, the rough appearance directly affected users' expectations of the overall product quality, even causing some to lower their evaluation before experiencing the functions.

5. User Suggestions

During the open discussion session, several participants gave specific suggestions. P1 hoped the appearance could be more integrated and aesthetically pleasing, with multiple virtual looks to choose from. P5 wanted more different expressions added. P3 wished the voice function were smarter and more reliable. Several participants also proactively asked whether pets could interact with each other. Although this feature was planned in the technical documentation, the current prototype did not implement it. Users' interest in this direction exceeded expectations.